

INTRODUCTION

Diagnostic yield in NGS genome diagnostics is constraint by the high fraction of variants of uncertain significance, in large part due to insufficient interpretability of missense variation. Existing pathogenicity predictors produce unexplainable scores. Here, we present the **Digital Approximation of Variant Effects (MOLGENIS DAVE)**, an explainable missense variant predictor built on biophysically grounded features such as:

- **Folding energy**
- **DNA/RNA/Ligand binding**
- **Changes in charge, hydrophobicity, isoelectric point**
- **Post-translational modifications (PTMs)**

RESULTS

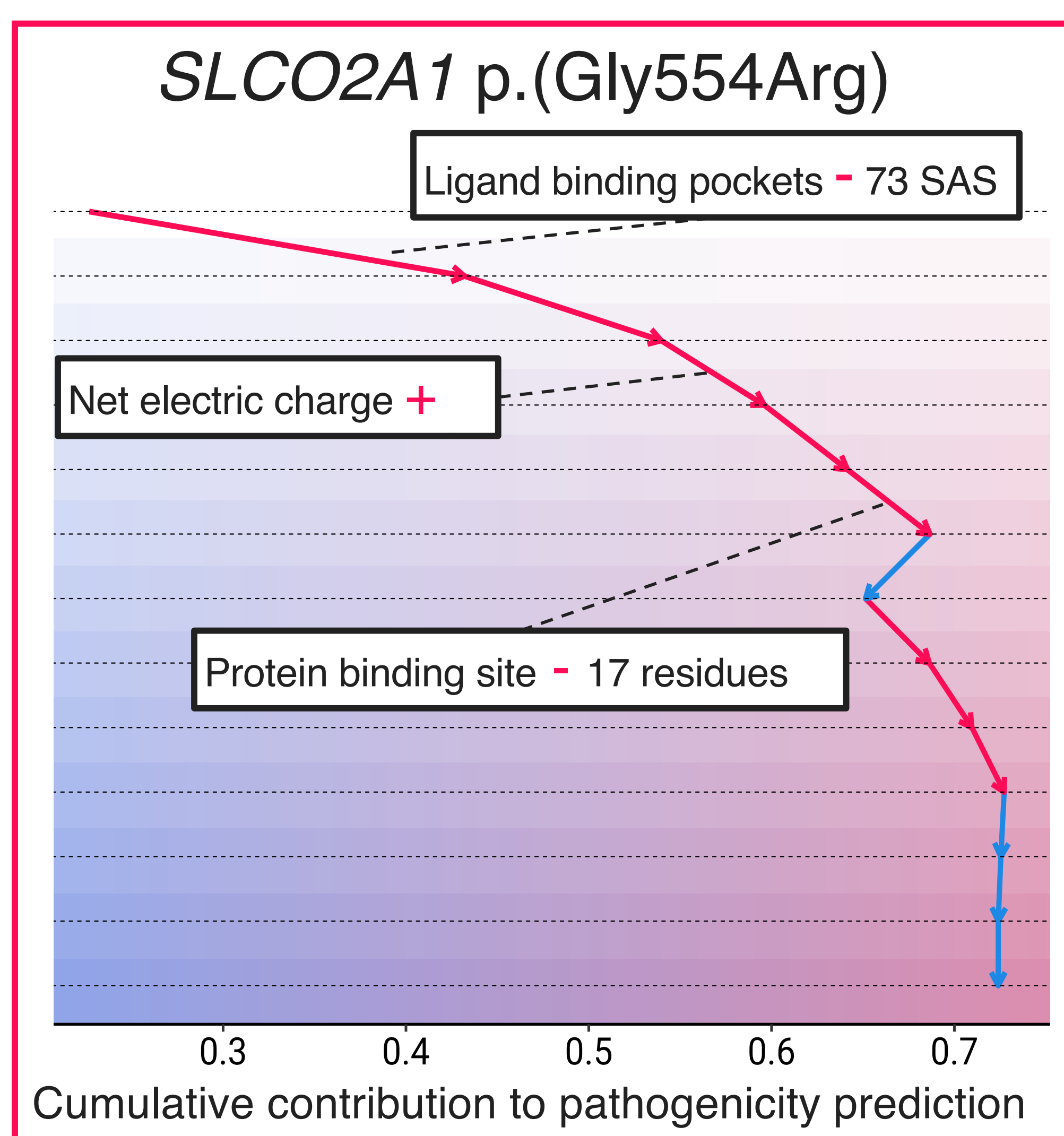


Figure 2: cumulative SHAP value plot

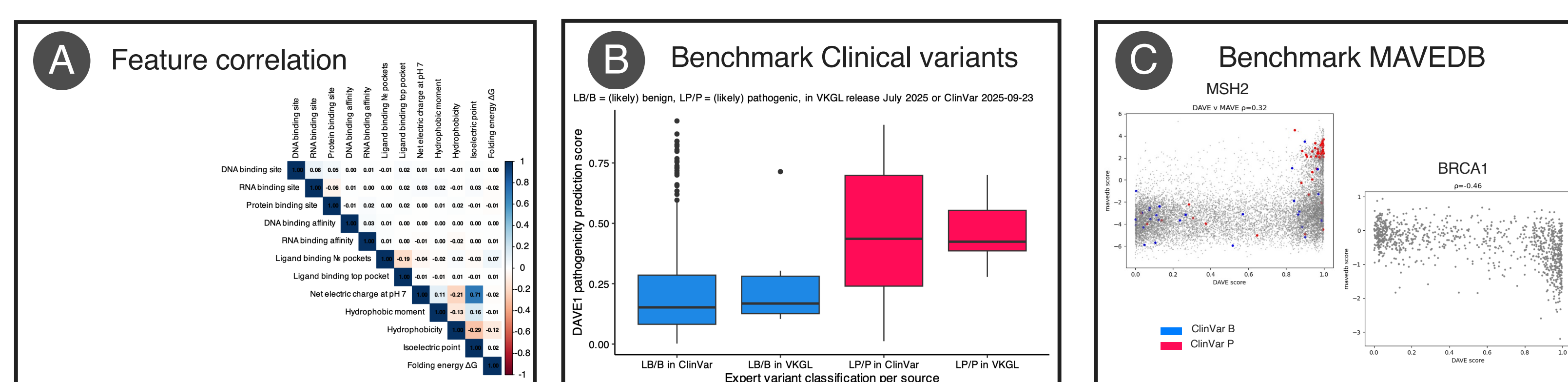


Figure 1: [a] Correlation of the features used in the DAVE model. [b] Benchmark of VUS variants with updated classifications in ClinVar and Dutch clinical variant dataset. [c] DAVE scores compared to scores from MAVEDB functional datasets.

The AUROC of the trained DAVE model on the testing set was **0.88**. Features were selected to correlate minimally (fig 1a), retrospective VKGL and ClinVar **VUS reclassification benchmark** (fig 1b), and MAVEDB functional data correlation (fig 1c).

Example of the DAVE model applied to a **VKGL VUS** variant in the **SLCO2A1** protein affecting the accessibility of a ligand pocket is shown in figures 2 and 3.

DISCUSSION

DAVE produces scores that can be interpreted mechanistically, making it a useful hypothesis-generation tool to guide targeted simulations and experimental validation. However, this interpretability comes at the cost of limited coverage of protein complexity, the use of only 12 features may overlook important aspects or context-dependent disease mechanisms. In contrast, models that incorporate evolutionary conservation often achieve higher predictive performance but in general provide no additional information in terms of molecular disease mechanism. Together, these observations highlight an important trade-off and motivate future work toward integrating high predictive accuracy with mechanistic interpretability in protein variant effect models.

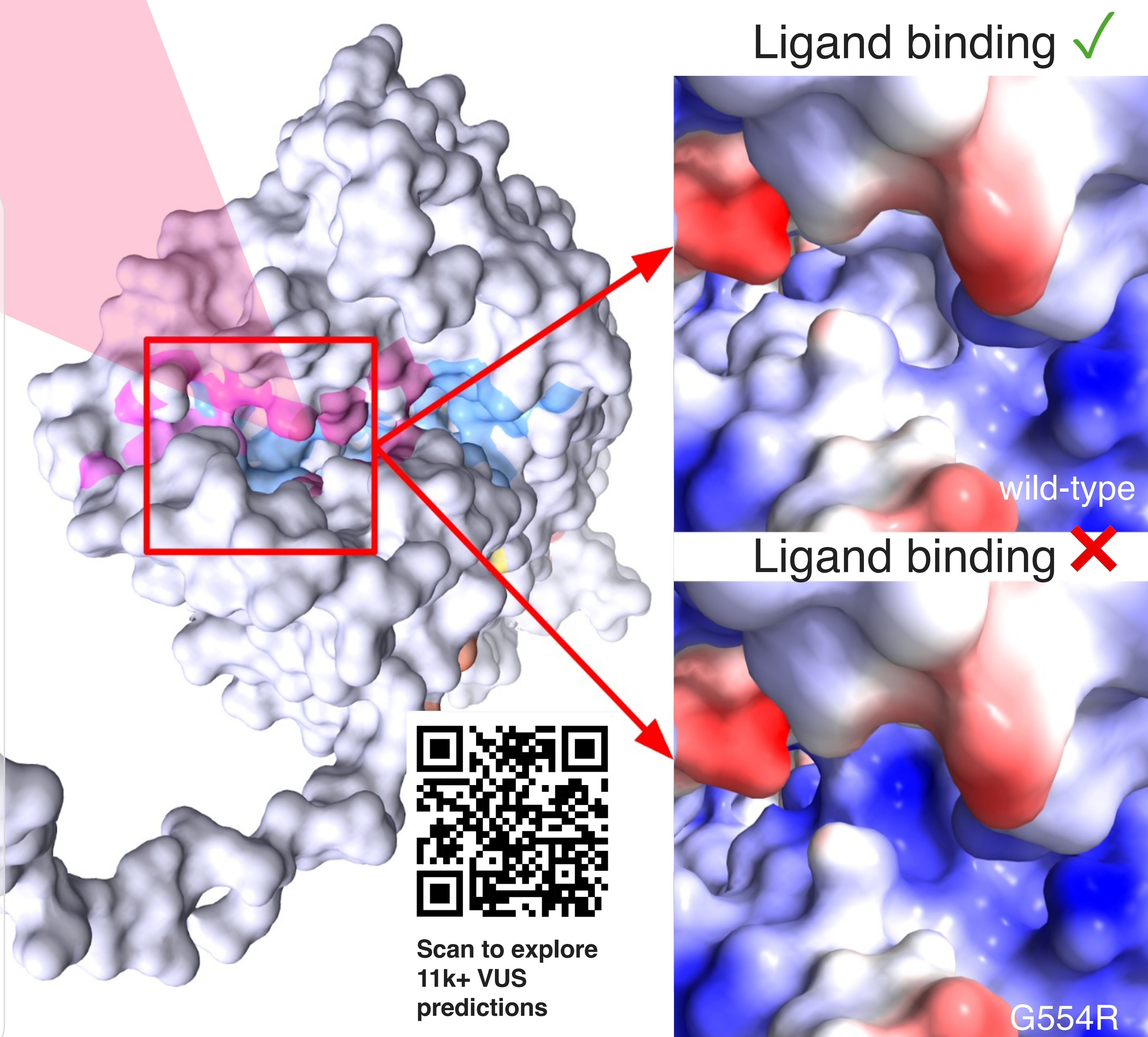


Figure 3: [right] ChimeraX protein visualization showing P2Rank predicted pockets, [left top] cropped electrostatic views of wild type protein and [left bottom] electrostatic view of the protein with the Gly554Arg substitution.